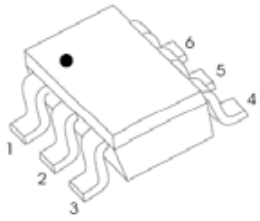
GOOD-ARK
SEMICONDUCTOR

GSBAT54xxW Series
Schottky Diode

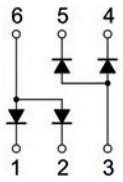
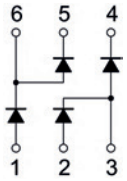
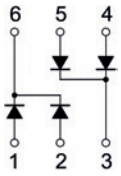
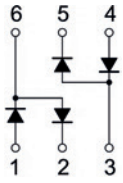
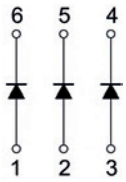
Features

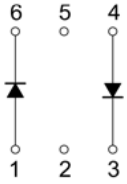
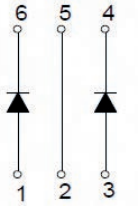
- Low forward voltage drop
- Fast switching
- PN junction guard ring for transient and ESD protection



SOT-363

Schematic Diagram

GSBAT54ADW	GSBAT54BRW	GSBAT54CDW	GSBAT54SDW	GSBAT54TW
				

GSBAT54DW	GSBAT54JW
	

Absolute Maximum Ratings (T_A=25°C unless otherwise noted)

Parameter	Symbol	Value	Unit
Repetitive Peak Reverse Voltage	V_{RRM}	30	V
Peak Working Reverse Voltage	V_{RWM}	30	V
DC Blocking Voltage	V_R	30	V
Forward Continuous Current	I_O	200	mA
Repetitive Peak Forward Current	I_{FRM}	300	mA
Non-Repetitive Peak Forward Surge Current @ t=8.3ms	I_{FSM}	600	mA
Power Dissipation	P_D	200	mW
Thermal Resistance From Junction To Ambient	$R_{\theta JA}$	500	°C/W
Junction Temperature	T_J	-55 to +125	°C
Storage Temperature	T_{STG}	-55 to +150	°C

Electrical Characteristics (T_A=25°C unless otherwise noted)

Parameter	Symbol	Test Conditions	Min	Max	Unit
Reverse Voltage	$V_{(BR)}$	I_R=100μA	30	-	V
Reverse Current	I_R	V_R=25V	-	2	μA
Forward Voltage	V_F	I_F=0.1mA	-	240	mV
		I_F=1mA	-	320	
		I_F=10mA	-	400	
		I_F=30mA	-	500	
		I_F=100mA	-	1000	
Total Capacitance	C_{tot}	V_R=1V, F=1MHz	-	10	pF
Reverse Recovery Time	t_{rr}	I_F=I_R=10mA, I_{rr}=0.1×I_R, R_L=100Ω	-	5	nS

Typical Characteristic

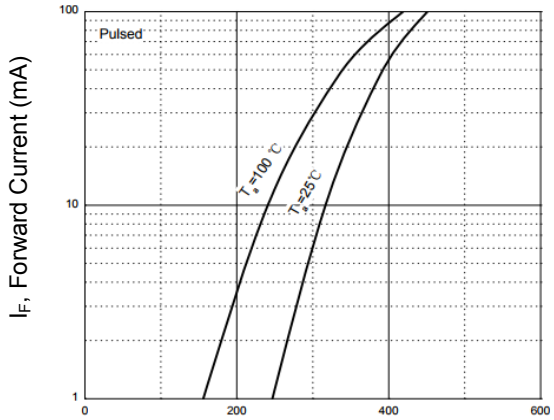


Figure 1. Forward Characteristics

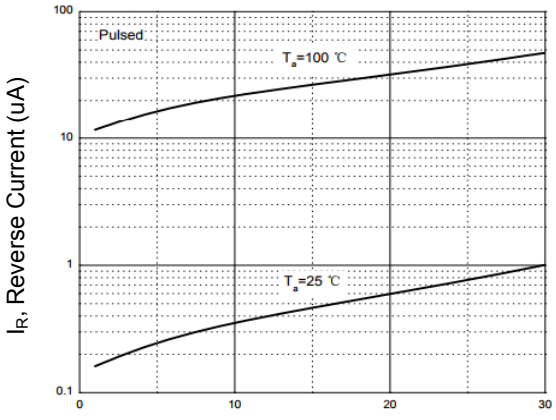


Figure 2. Reverse Characteristics

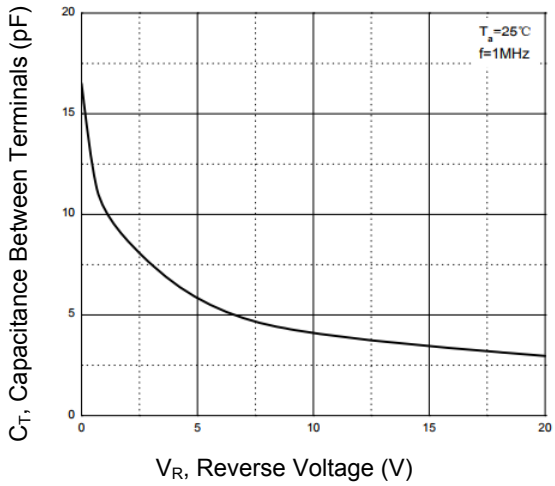


Figure 3. Capacitance Characteristics

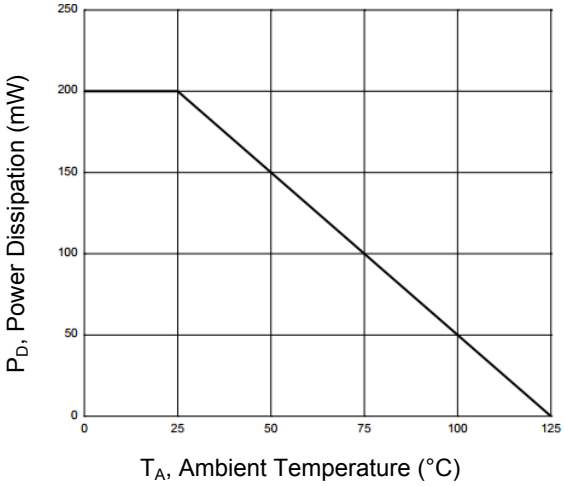
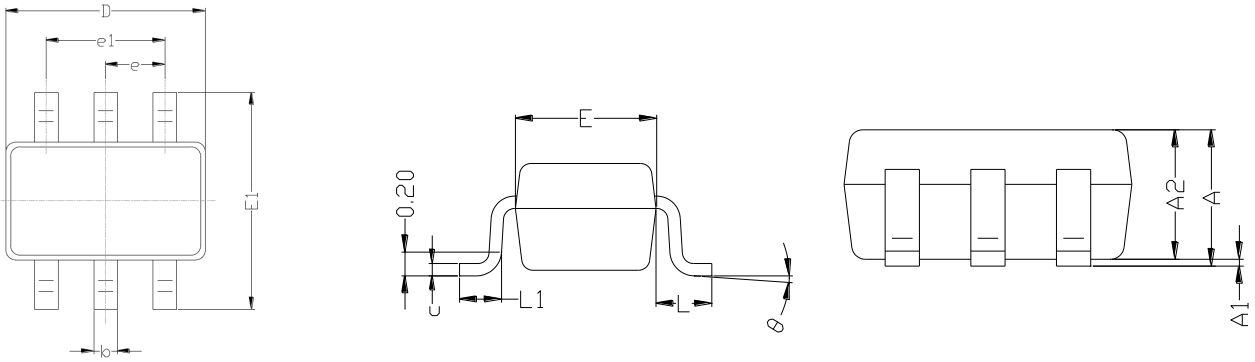


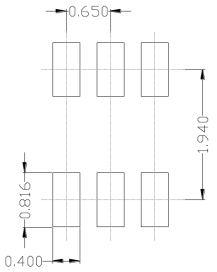
Figure 4. Power Derating Curve

Package Outline Dimensions (SOT-363)



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	0.900	1.000	0.035	0.039
A1	0.000	0.100	0.000	0.004
A2	0.900	1.000	0.035	0.039
b	0.150	0.350	0.006	0.014
c	0.100	0.150	0.005	0.006
D	2.000	2.200	0.079	0.087
E	1.150	1.350	0.045	0.053
E1	2.150	2.450	0.085	0.096
e	0.650 TYP		0.026 TYP	
e1	1.200	1.400	0.047	0.055
L	0.525 REF		0.021 REF	
L1	0.260	0.460	0.010	0.018
θ	0°	8°	0°	8°

Recommended Pad Layout



Note:
1. Controlling dimension: in millimeters.
2. General tolerance: ±0.05mm.
3. The pad layout is for reference purposes only.

Order Information

Device	Package	Marking	Carrier	Quantity
GSBAT54ADW	SOT-363	KL6	Tape & Reel	3,000pcs / Reel
GSBAT54BRW	SOT-363	KLB	Tape & Reel	3,000pcs / Reel
GSBAT54CDW	SOT-363	KL7	Tape & Reel	3,000pcs / Reel
GSBAT54SDW	SOT-363	KL8	Tape & Reel	3,000pcs / Reel
GSBAT54TW	SOT-363	KLA	Tape & Reel	3,000pcs / Reel
GSBAT54DW	SOT-363	KLD	Tape & Reel	3,000pcs / Reel
GSBAT54JW	SOT-363	KLC	Tape & Reel	3,000pcs / Reel